

Summary

AI Engineer with a strong foundation in building LLM-based applications and Retrieval-Augmented Generation (RAG) systems. Experienced in implementing Generative AI workflows using LangChain, HuggingFace, and Groq APIs. Skilled in Python-based development, model evaluation, and integrating vector stores like FAISS and ChromaDB. Previously worked as a Business Analyst, bringing a structured problem-solving mindset to AI system design. Actively upskilling in MLOps, LLM lifecycle, and scalable AI deployment.

Skills

Programming: Python, SQL (SQL Server, PostgreSQL, MySQL)

Machine Learning: Scikit-learn, Feature Engineering, TensorFlow

Generative AI: LangChain(AI Agents), LangGraph (Agentic Workflows), RAG Pipelines, LLM Fine-Tuning, Prompt Engineering

LLMs & Vector DBs: GPT-4, Llama 3, Hugging Face Transformers, Groq, ChromaDB, FAISS, Hybrid Search and embeddings

Backend & Evaluation: FastAPI, Streamlit, RAGAS, Comet Opik(Tracing), MLflow

Deployment & DevOps: Docker, Git, Azure OpenAI Services, Azure Cloud ,CI/CD - GitHub Actions

Projects

RAG_SYSTEMS: Exploratory RAG Implementations

April 2025 – Present

- **Tools/Technologies:** Python,pydantic, LangChain, Groq, FAISS, Ragas, HuggingFace Transformers, ChromaDB, Unstructured.io, Docker, Streamlit, FastAPI, Comet ML, Comet Opik, Cohere, Pandas, NumPy, Git, Ollama, Docker .
- Developed a modular repository of Retrieval-Augmented Generation (RAG) implementations covering key components such as multi-modal pipelines, reranking, memory handling, and observability
- Includes submodules like RAG Agents, Finetuning strategies, and Evaluation workflows; designed for experimentation and rapid prototyping using LangChain and LLM APIs
- Built a RAG pipeline using LangChain to extract and answer questions from text, tables, and images in PDFs using Unstructured.io, HuggingFace embeddings, and ChromaDB

Finetuned Chatbot: Sarcastic Chandler Bing AI

July 2025 – Present

- Developed and fine-tuned a custom chatbot to embody the sarcastic personality of Chandler Bing, leveraging a Hugging Face dataset, Unsloth, and LoRA for efficient parameter-efficient fine-tuning of an LLM.
- Engineered a full-stack chatbot solution, including data preparation pipelines, a FastAPI server for API interaction, and a Streamlit-based interactive UI, enabling local deployment via Ollama.

Kaggle competition - Exploring Mental Health Data

November 2024

- Analyzed a large dataset of mental health survey responses using Python libraries (pandas, NumPy)
- Identified key factors contributing to depression and engineered novel features to improve model performance
- Developed and implemented machine learning models (CatBoost) with hyperparameter optimization (Optuna) to predict depression likelihood
- Achieved 94 percent accuracy and improved F1-score by 2 percent through feature engineering, leading to better precision and recall in depression prediction

Up-skilling / Career Transition

Independent AI Engineering Study & Project Development

August 2024 - Present

- Pursued independent study in AI/ML engineering with focus on LLM lifecycle, RAG systems, and Python problem-solving
- Built and deployed end-to-end Generative AI applications including multi-modal RAG systems; earned AWS Certified AI Practitioner

Experience

Business Analyst - Novac Technology Solution

August 2022 - August 2024

- Collaborated with development teams and stakeholders to gather and define business requirements for data-driven and analytical projects, ensuring alignment with strategic objectives
- Utilized SQL for data analysis and querying, ensuring data accuracy and integrity
- Conducted manual testing to identify and resolve defects, improving project quality
- Contributed to achieving CMMI Level 5 certification by streamlining data management processes

Education

M.B.A. in Finance and Marketing

2019 - 2021

Bharathiar University, Coimbatore | CGPA: 7.00/10.00

B.E. in Mechanical Engineering

2015 - 2019

Anna University, Erode | CGPA: 7.28/10.00

Certifications

AWS Certified AI Practitioner (AIF-C01) Amazon Web Services (AWS)

March 2025